

# Prospective Learning Needs Physical Implementations

## *Phase-Structured State Dynamics as a Discriminating Neural Hypothesis*

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### **Abstract**

Prospective learning asks how an agent should adapt when environments and its own capabilities change. That agenda identifies a problem, not a biological implementation. The immediate physical problem is that a nervous system must use its existing structure and ongoing activity to change behavior before slow structural learning has rewritten the network. This paper first explains that operation in ordinary language and only then names a Self-Aware Networks (SAN) hypothesis: discrete spikes occur within continuous population dynamics; arriving activity is interpreted relative to a receiver's current state and expected timing; phase-compatible events can alter transfer and route selection; and the resulting recurrent update may reduce a local mismatch while preserving distinctions elsewhere. Phase Wave Differential (PWD), Neural Array Projection Oscillation Tomography (NAPOT), difference dissipation, conditional reset amplification, Activative Memory, and Totality are used as names for parts or consequences of that proposed operation, not as substitutes for an explanation.

Three neighboring claims are kept separate. Prospective learning is a future-facing objective; prospective configuration is activity-before-plasticity in specified energy-based model families; and mesoscopic phase patterns are measurable in specified cortical preparations. No cited source establishes their conjunction. Stage 0 therefore asks whether the proposed analysis can recover planted mechanisms in inspectable synthetic networks. The implemented deterministic benchmark calibrated thresholds on 128 networks and applied them unchanged to 192 held-out networks. It recovered all 48 reference phase-gated systems, rejected all 144 reference nonphase systems, and localized preferred phase with 0.074-radian median error. Adversarial tests bound that result: sensitivity fell to 8.3 percent for weak planted effects and 12.5 percent under elevated noise, while a phase-locked intervention artifact produced a 2.8 percent false-positive rate.

A two-stage biological protocol is then specified without claiming that it has been performed. Stage A asks whether engineered phase geometry improves next-outcome prediction beyond a locked nonlinear raw-waveform or recurrent champion. Stage B asks whether matched perturbation at separately localized open versus closed beta phase changes a predeclared receiver-transfer endpoint. A deterministic clustered simulation stress-tests declared smallest effects across hypothetical nuisance profiles; it cannot select an animal or session count before blinded pilot estimates, hardware characterization, ethical review, and an approved maximum exist. No biological data, pilot estimate, achieved power, completed preregistration, or biological causal result is reported.

## **Scope and evidence boundary**

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## **1. An Objective Is Not Yet An Implementation**

Kording, Vogelstein, Chaudhari, and Verstynen argue that intelligence must anticipate how future environments and organismal capabilities evolve. Their 2025 Neuron article extends a prospective learning program developed in earlier theoretical work. The program concerns objectives, distributional change, planning, benchmark design, and decision quality. It does not commit to one neural code.

A biological implementation must explain at least four operations:

1. represent a current state and a distribution over near-future states;
2. update that representation when transition structure changes;
3. select behavior before extensive retraining; and
4. consolidate useful rapid states into slower plastic change without catastrophic interference.

Many recurrent systems can perform these operations in principle. A phase account earns scientific content only if it predicts something beyond recurrent and nonlinear raw-input models supplied with the same underlying observations.

The sources play different evidentiary roles. Kording et al. define the prospective-learning agenda; De Silva et al. formalize a future-facing learning problem; Song et al. define prospective configuration in energy-based models; and Denker et al. and Bhattacharya et al. establish phase-pattern measurements in motor and prefrontal preparations. Moving support from one role to another would create a new hypothesis rather than a citation shortcut. The release supplement includes a claim-to-source matrix and exact locator ledger so these roles can be audited independently.

## 2. Earn The Mechanism Before Naming It

Consider an animal that has learned that a visual cue predicts one outcome, but the cue-action-outcome relation changes without warning. The sensory cue and available movement can remain the same while the useful response changes. The nervous system must therefore combine current evidence, recent outcomes, existing synaptic structure, ongoing population activity, and its own action history; update which route is favored; act before extensive structural rewriting; and allow slower plasticity to consolidate useful states.

Established neuroscience supplies several components of this problem: recurrent activity can preserve and transform state; dendritic and inhibitory conditions affect how inputs are integrated; oscillatory phase can alter communication opportunities; and synaptic plasticity changes future accessibility. These facts do not by themselves establish the joined SAN mechanism. The proposed operation is more specific:

1. learned dendritic and synaptic structure and ongoing population activity jointly establish a receiver's current context;
2. discrete spikes arrive within continuous, time-varying activity rather than on an empty channel;
3. each receiver transforms an event according to its own state, conduction delay, inhibition, amplitude, and expected relative timing;
4. sufficiently compatible arrivals can sum, gate transfer, recruit or reset a population, and change which downstream route gains influence;
5. recurrent consequences return as new sensory, motor, and internal evidence; and
6. fast state change can be consolidated into slower plastic change if the resulting route remains useful.

Only after that operation is understood are the SAN terms introduced below.

### 2.1 Candidate state variables

For node  $i$  and band  $\omega$ , write analytic activity as

$$x_i^\omega(t) = a_i^\omega(t) \exp(i\phi_i^\omega(t)). \quad [1]$$

A phase-structured state can include:

- relative phase  $\Delta\phi_{ij}$  rather than absolute phase;
- spatial phase gradient and its temporal stability;

- amplitude and duration as separate variables;
- cross-frequency timing;
- dendritic and inhibitory state;
- network geometry and conduction delay; and
- the recent sequence of actions, sensory outcomes, and prediction errors.

Tonic and phasic are relative roles, not permanent frequency labels. A tonic component is the locally persistent context for the scale and task being analyzed. A phasic component is an update relative to that context. Gamma can be tonic in a brief active state; theta can be phasic relative to a slower baseline.

SAN's stronger hypothesis is that learned dendritic and synaptic structure and transient population patterns are alternating physical forms of a state. Fast activity expresses or tests a candidate future configuration; slower plasticity changes which configurations become easy to re-enter. This is compatible with multiple learning rules and does not make phase the only code.

## 2.2 Receiver-relative difference: Phase Wave Differential

The relevant signal is not assumed to be an isolated spike or an absolute phase. An arriving event can have different consequences depending on the receiver's ongoing state and the relation expected along that route. SAN calls the measurable departure among the arriving event, the receiver's present state, and the expected delayed relation a **Phase Wave Differential (PWD)**. A spike at a particular phase samples a continuous trajectory; it does not literally contain the entire trajectory. Repeated receiver-relative samples may, however, supply information about the rate and direction of change.

PWD is therefore not another name for phase coding, predictive coding, or a traveling wave. Those established ideas concern, respectively, information in phase relations, prediction-error organization, or propagating population activity. PWD adds an explicitly receiver-relative transformation: what arrived, what state received it, what relation was expected after delay, and what downstream change followed. Stage A tests whether that added representation is useful beyond raw-waveform and recurrent comparators; Stage B tests whether the localized phase relation has a selective causal consequence.

## 2.3 Coincidence, route selection, and conditional reset

When multiple inputs arrive within a receiver's integration window, their effects can sum or cancel according to synaptic sign, timing, membrane state, and inhibition. SAN's coincidence proposal treats phase-compatible arrivals as a candidate mechanism for making one route more effective than another. This does not imply that a single spike is always noise, that all useful signals must synchronize globally, or that neural coherence supplies muscular energy.

The stronger **conditional reset amplification** hypothesis is testable: a small timing-sensitive event may recruit inhibition or another reset process that changes a much larger population's phase organization. The older SAN source proposed a transition from noise-like square-root scaling to coherent linear scaling. This preprint does not assume that scaling law. It instead asks whether response magnitude, phase reorganization, and receiver transfer grow nonlinearly after a separately measured reset threshold and whether matched nonphase perturbations fail to produce the same effect.

## 2.4 Local mismatch reduction: difference dissipation

If a receiver-relative mismatch continues to influence recurrent activity and plasticity, later states may reduce that same local residual. SAN calls this candidate contractive update **difference dissipation**. The term does not mean that information or energy vanishes, and it does not require all neural activity to converge on

global synchrony. Local mismatch can decrease while new distinctions, splay states, inhibition, or prediction errors remain informative elsewhere. Any thermodynamic interpretation requires separate measurements of metabolic cost and cannot be inferred from phase alignment alone.

## 2.5 Receive, transform, project, and re-enter

Different populations hold partial conditions rather than a complete picture. Each receiver transforms what reaches it; sends an updated state onward; and receives returned consequences after network, body, or environmental change. SAN calls this distributed receive-transform-project-reenter cycle **Neural Array Projection Oscillation Tomography (NAPOT)**. “Tomography” and “scanning” are functional analogies to reconstruction from partial, differently situated measurements, not claims that a literal scanner or hidden viewer exists in the brain. The multiple-aperture sensor analogy similarly motivates computational alignment across offsets; it is not biological evidence.

At a larger architectural scale, **Activative Memory** names the presently recruited subset of learned structure, while **Totality** names the adaptive world-and-self model distributed across available structure and activity. These terms describe possible consequences of the lower-level operator. The primary tests in this paper concern receiver-relative representation and transfer; they do not make consciousness, selfhood, or a global standing wave an endpoint.

## 3. Separate Three Neighboring Ideas

### 3.1 Prospective learning

Prospective learning is the task-level objective: learn for a future whose distributions, goals, or capabilities may change.

### 3.2 Prospective configuration

Prospective configuration is Song et al.’s computational activity-before-plasticity principle: in the studied model families, activity first moves toward a target-compatible configuration and synaptic change later consolidates that state. It is a genuine comparator, not yet an established in-vivo biological mechanism. SAN does not receive priority merely because both programs use the word prospective.

### 3.3 Phase-structured implementation

The SAN residual is narrower and more physical: receiver-relative multiscale phase geometry, dendritic coincidence, and conditional reset may supply a useful representation or inductive bias that locked non-oscillatory and nonlinear raw-input models do not recover as efficiently or robustly.

The three layers can coexist. A prospective-learning benchmark could be solved by prospective configuration with or without a phase-specific contribution.

Doerig et al.’s neuroconnectionist framework adds a useful standard without settling the mechanism: a model family earns value by generating explanatory and predictive progress. That standard is compatible with testing a SAN-derived phase representation, but it does not privilege phase, recurrence, or any other implementation in advance.

## 4. Design Status And No-Result Boundary

The remainder of this article is a prospective design freeze, version KORDING-D3-DF-001. It is not a completed or registered study. The task probabilities, model budgets, smallest effects of interest, confidence-interval rules, and phase-error bound below are proposed commitments. They are not measurements, pilot estimates, power outputs, or evidence that the effects exist.

No animal count, session count, transition count, stimulation current, pulse width, or charge density is supplied here. Those quantities require blinded nuisance estimates, hardware characterization, ethical review, and a documented design simulation. If the ethically available maximum cannot meet the declared precision rule, Stage B must not be represented as a confirmatory causal test.

## 5. Stage 0: Calibrate The Analysis Before Interpreting Biology

Lindsay and Bau argue that systems-neuroscience tools should be tested in artificial networks whose complete state and mechanism can be inspected. That proposal directly strengthens this protocol. A neural-analysis pipeline that cannot recover a known planted phase gate, or that falsely reports one in a phase-free network, is not ready to identify a biological implementation.

Stage 0 therefore precedes access to the biological confirmatory folds. It uses four task-matched recurrent systems trained on the same regime-switch problem:

| Ground-truth system | Planted mechanism                                                                                  | Required diagnostic                                                             |
|---------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| G_recurrent         | recurrence and nonlinear state only                                                                | phase pipeline must not invent a phase-specific mechanism                       |
| G_phase             | explicit receiver-relative phase gate controlling a known transfer edge                            | pipeline must localize the gate and recover its direction                       |
| G_amplitude         | amplitude- or burst-gated transfer without phase selectivity                                       | pipeline must distinguish amplitude gating from phase gating                    |
| G_delay             | conduction-delay and recurrent dynamics that create phase-correlated activity without a phase gate | predictive phase features may appear, but causal attribution to phase must fail |

### 5.1 Implemented reference calibration

The deterministic reference benchmark applies the same feature construction, held-out prediction, phase localization, and matched-intervention analysis to shuffled network identities. A 128-network calibration partition, 32 networks per mechanism, selected thresholds from a declared grid. The frozen rule requires all three conditions in both task contexts: at least 0.08 relative held-out MSE gain from the phase-augmented model, at least 2.0 standardized open-minus-closed transfer effect, and a positive causal direction. Ground-truth labels are used only after network-level outputs have been computed.

The unchanged rule was then evaluated on 192 held-out networks, 48 per mechanism:

| Held-out measure                              | Reference result |
|-----------------------------------------------|------------------|
| G_phase sensitivity                           | 1.000            |
| Nonphase specificity                          | 1.000            |
| False-positive rate                           | 0.000            |
| Positive direction replicated across contexts | 1.000            |
| Median preferred-phase localization error     | 0.0743 radians   |

The reference generator family therefore passes this bounded implementation check. This is code calibration against planted synthetic mechanisms, not evidence that the same mechanism exists in a biological network.

## 5.2 Prediction-causation firewall

The strong amplitude-phase-correlation stress test demonstrates why both parts of the rule are necessary. For `G_amplitude`, median held-out predictive phase gain was 0.228, above the frozen predictive threshold, even though the generator contained no phase-selective transfer gate. Its phase-mechanism classification rate remained zero because the matched causal contrast was absent. Predictive phase structure alone therefore does not receive a mechanistic interpretation.

## 5.3 Adversarial limits

| Stress condition                            | Sensitivity | False-positive rate | Bound exposed                                            |
|---------------------------------------------|-------------|---------------------|----------------------------------------------------------|
| Weak planted phase effect                   | 0.083       | 0.000               | conservative threshold misses weak gates                 |
| Elevated observation and intervention noise | 0.125       | 0.000               | sensitivity collapses before specificity                 |
| Strong amplitude-phase correlation          | 1.000       | 0.000               | causal requirement rejects a predictive confound         |
| Phase-locked intervention artifact          | 1.000       | 0.028               | invalid pathway specificity can create false attribution |

These failures are not implementation defects to conceal. They define the current operating envelope. Biological use still requires independent methods review, a frozen simulator / error budget, pilot-derived nuisance estimates, validated target engagement, and negative controls capable of detecting phase-locked sensory or pathway artifacts.

This stage also addresses a limit emphasized by Kording-linked causal work: recorded activity can predict an outcome while downstream transformations make that activity a poor guide to causal contribution. The inspectable systems permit exhaustive unit and pathway interventions, so the proposed phase estimators can be compared with known causal effects rather than only with decoding accuracy.

## 6. Frozen Dynamic-Transition Preparation

The primary preparation is a prospective, within-animal, multisession macaque lateral-prefrontal cortex study using simultaneously recorded spikes and local field potentials from a candidate sender and receiver population. This choice fixes a preparation for evaluation; it does not claim that an experiment has begun or that a specific site is safe or available.

### 6.1 Regime-switch task

Each valid trial presents one of four visual cues, permits one of two saccadic actions, and returns one of two visually distinct outcome tokens. The sensory set, action set, timing, and reward scale remain constant. Only the latent cue-action-outcome transition matrix changes.

- In regime R0, each cue has one dominant action-outcome mapping with probability 0.80; the alternative occurs with probability 0.20.
- In regime R1, the dominant and alternative mappings are exchanged.

- Regime dwell length is drawn before the session from a discrete uniform distribution over 60 to 100 valid trials.
- Switches are not announced to the animal and do not introduce a novel cue, motor requirement, or reward magnitude.
- A transition episode comprises the final 20 valid trials before a switch and the first 20 valid trials after it. Transition episodes are the indivisible resampling and cross-validation unit.

The primary prediction time is action commitment and precedes outcome delivery. The available history contains the current cue and chosen action, all prior cues, actions, outcomes, and prediction errors, and the preceding 500 ms of declared neural input. The primary target is the probability of the next outcome token. The primary loss is held-out negative log likelihood in nats per valid trial.

## 6.2 Independent localization

Before opening either Stage A test folds or Stage B condition labels, a separate localization block must freeze:

1. one sender-receiver pair in lateral prefrontal cortex;
2. one beta center frequency constrained to the 15-30 Hz band;
3. one receiver reference channel and rereferencing rule;
4. a past-only phase estimator and its filter order;
5. one receiver-transfer measurement window of 10-80 ms after perturbation; and
6. one site-specific open phase  $\theta_{\text{open}}$  estimated only from localization data.

The closed phase is defined mechanically as  $\theta_{\text{open}} + \pi \bmod 2\pi$ . The open phase is not re-estimated after viewing causal outcomes. A held-out causal replay must show maximum circular targeting error no greater than  $\pi/8$  after measured acquisition, estimation, command, and device latencies are included. A site that fails this engineering criterion is excluded before condition outcomes are viewed.

## 7. Stage A: Representation-Utility Test

All models receive the same behavioral history and the same underlying neural recording interval. Engineered phase variables are deterministic transforms of that recording. Therefore a positive Stage A result can establish representation utility under the declared model classes, not information absent from the raw signal and not causal necessity.

This distinction is a primary causal-identification boundary. High held-out performance can show that a variable is useful to a declared predictor; it cannot show that the variable carries the intervention target used by the network. Stage A must therefore report prediction, calibration, and out-of-context adaptation without using words such as mechanism, control, or necessity unless Stage 0 and Stage B independently support them.

### 7.1 Comparator ladder

| Model  | Locked input and role                                                               |
|--------|-------------------------------------------------------------------------------------|
| M_hist | cue, action, outcome, reward, and trial-history baseline                            |
| M_rate | M_hist plus binned spike rate, beta power, broadband power, and aperiodic summaries |
| M_dyn  | recurrent state model using M_rate inputs without engineered phase geometry         |

| Model   | Locked input and role                                                                                                                                                            |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M_raw   | nonlinear dilated temporal-convolution model receiving the raw multichannel waveform, spikes, and behavioral history; it can learn filters, interactions, and temporal structure |
| M_phase | the same underlying raw inputs and recurrent prediction head augmented with declared receiver-relative phase, phase-gradient, duration, and cross-frequency features             |

Within each outer training fold, inner validation selects one champion from the preregistered M\_dyn and M\_raw families. That selected champion is then evaluated once on the outer test fold; outer-test performance cannot influence family or hyperparameter selection. M\_raw cannot be discarded because it is the direct test of whether a generic nonlinear learner can recover the claimed utility from raw input without hand-supplied phase features.

## 7.2 Data and compute lock

- Outer folds hold out complete transition episodes; no trial from a held-out episode may enter feature fitting, normalization, model selection, early stopping, or imputation.
- All preprocessing choices, including filters and reference transforms, are fit inside the training partition.
- Each model family receives 24 hyperparameter evaluations, at most 100,000 optimizer updates per evaluation, the same early-stopping patience, and the same five initialization seeds.
- Trainable parameter counts for M\_raw and M\_phase must be within 5 percent. The accelerator type, numerical precision, and maximum training examples are identical.
- The result is reported for every seed and outer fold; selection of a favorable seed is forbidden.
- Phase-scrambled, block-preserving circular-shift, task-irrelevant phase, reference-sensitivity, and preprocessing-sensitivity controls use the same outer folds.

Equal compute and similar parameter count do not prove equal representational capacity. They define the bounded comparison under which a representation-utility claim is allowed.

## 7.3 Stage A estimand and SESOI

Let  $L_{\text{champion}}$  be held-out log loss from the better training-fold-selected M\_dyn or M\_raw family, and let  $L_{\text{phase}}$  be the held-out phase-model loss. The episode-level estimand is

$$\Delta_A = L_{\text{champion}} - L_{\text{phase}}. \quad [2]$$

Positive values favor the engineered phase representation. The Stage A smallest effect size of interest is fixed prospectively at

$$\delta_A = 0.02 \text{ nats per valid trial}. \quad [3]$$

This is a design threshold, not an estimated effect. A 0.02-nat average improvement corresponds to a geometric-mean likelihood ratio of  $\exp(0.02)$ , approximately 1.020, for the realized outcome. The threshold demands a repeatable gain before the added representation is treated as practically useful. Secondary outcomes are calibration error, adaptation trials to a frozen behavioral criterion, and cumulative regret; they cannot rescue a failed primary result.



## 8. Precision And Sample-Size Rule

Stage A and Stage B receive separate design simulations. Before outcomes are unblinded, nuisance parameters may be estimated only from condition-pooled or label-scrambled data: valid-trial rate, transition-episode yield, within-session variance, between-session variance, between-animal variance, phase-lock success rate, and missingness. No open-versus-closed or phase-versus-champion effect estimate may enter this step.

For each stage, simulate the complete hierarchical analysis at the corresponding SESOI. The chosen animal, session, and transition-episode counts must satisfy both conditions in at least 90 percent of simulated datasets:

1. the two-sided 95 percent confidence interval excludes zero in the predicted direction; and
2. the 95 percent confidence-interval half-width is no greater than one half of the SESOI.

The simulation code, random seed, variance inputs, attrition assumptions, ethical maximum, and resulting counts must be frozen before confirmatory labels are opened. There is no outcome-driven optional stopping. A preregistered attrition replacement rule may restore the frozen analyzable count, but cannot increase it after inspecting the effect.

### 8.1 Hypothetical nuisance sensitivity

The release supplement includes an executable stress test rather than a design calculation. It simulates clustered animal means for four hypothetical designs and three declared variance-and-attrition profiles. None of the nuisance values comes from pilot data. The simulation uses the declared SESOIs and the joint success rule: the 95 percent interval must exclude zero in the predicted direction and its half-width must not exceed one half of the SESOI in at least 90 percent of datasets.

Under the optimistic profile, hypothetical S2 clears the rule in 94 percent of Stage A simulations but only 63 percent of Stage B simulations. Under the reference profile, success reaches 99.25 percent for Stage A and 92.5 percent for Stage B only at the largest hypothetical S4 design. Under the adverse profile, even S4 reaches only 34 percent for Stage A and 16.25 percent for Stage B. Across all null cells, the largest false-positive-and-precision success rate is 3 percent.

These numbers are sensitivity results under invented stress-test inputs. They do not recommend S4, establish feasibility, provide achieved power, or justify animal use. Their result is methodological: the study cannot freeze counts from the SESOI alone. It needs blinded estimates of variance, yield, phase-lock success, missingness, and hardware failure plus an independently approved ethical maximum. If no design below that maximum clears the rule under a conservative nuisance range, Stage B should not proceed.

## 9. Stage B: Phase-Specific Causal Test

Stage B is logically independent of the Stage A decoder. It requires separate ethical, safety, engineering, and precision approval. The randomized contrast identifies only the effect of assignment to the declared open-versus-closed targeting rule under the following claim contract:

| Assumption        | Operational check                                                                                            | Failure consequence            |
|-------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------|
| Target engagement | held-out replay meets the $\pi/8$ phase-error bound and delivered-field telemetry matches the frozen command | no phase-specific causal claim |

| Assumption             | Operational check                                                                                   | Failure consequence                                                |
|------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Dose consistency       | waveform, charge, timing opportunity, and hardware route match across active conditions             | contrast is confounded by dose                                     |
| Limited carryover      | prespecified washout and lag diagnostics exclude material influence on the next randomized episode  | estimate is not an episode-local effect                            |
| Bounded interference   | transition episodes are the randomization unit and simultaneous cross-site conditions are forbidden | ordinary unit-level causal interpretation is blocked               |
| Missingness discipline | hardware and artifact failures use label-blind rules with declared sensitivity analyses             | complete-case result is not confirmatory                           |
| Pathway specificity    | active phase-averaged and sham controls bound nonspecific stimulation and sensation                 | open-versus-closed difference cannot identify the proposed pathway |

Transition episodes are randomized in balanced blocks among:

1. predicted open phase  $\theta_{\text{open}}$ ;
2. predicted closed phase  $\theta_{\text{open}} + \pi$ ;
3. active phase-averaged control with target phase drawn uniformly before the trial; and
4. sham with identical handling and no delivered current.

On each of the first four valid post-switch trials, the active conditions deliver one charge-balanced biphasic pulse pair at the first qualifying beta cycle after action commitment. Waveform, current, pulse width, interphase interval, pulse count, total charge, transition timing, and device route are identical across active conditions. The site-specific numeric dose is set in a separate blinded safety and hardware calibration and then frozen. It is intentionally absent here; no current or charge-density value may be invented before that review.

The past-only online estimator, measured end-to-end latency correction, and  $\pi/8$  replay criterion are part of the causal lock. Trials with prospectively defined estimator or hardware failure are flagged without viewing the condition outcome. Movement, local beta power, baseline receiver firing, and stimulation artifact metrics are recorded as prespecified diagnostics, not used to redefine the preferred phase.

## 9.1 Primary causal endpoint and SESOI

Receiver transfer is the artifact-cleaned receiver spike-count or evoked-response change in the fixed 10-80 ms window relative to its matched prestimulus expectation. It is standardized by a condition-pooled residual standard deviation estimated from the blinded calibration data. The primary causal estimand is

$$\Delta_B = E[T_z | \text{open}] - E[T_z | \text{closed}]. \quad [4]$$

The Stage B smallest effect size of interest is

$$\Delta_B = 0.20 \text{ residual-standard-deviation units}. \quad [5]$$

This threshold is a prospective judgment that a smaller receiver-transfer difference would not justify a phase-specific invasive claim. It is not a pilot effect. The primary analysis is a hierarchical model with fixed perturbation condition and prespecified baseline covariates, random intercepts for animal, session, and sender-receiver site, and a random open-versus-closed slope where the realized design supports it. Behavioral adaptation is secondary and cannot replace the receiver endpoint.

Behavioral grounding is a separate endpoint, not an automatic consequence of a receiver-transfer effect. Following Lindsay’s behavior-grounding criterion, Draft 5 prespecifies a secondary mediation ladder:

1. assignment changes receiver transfer;
2. receiver-transfer change predicts adaptation under held-out conditions;
3. the assignment-to-behavior effect is attenuated when the frozen receiver mediator is included; and
4. the sign and ordering replicate in context B.

This ladder is exploratory unless a later preregistration supplies a measurement model, temporal ordering, mediator-outcome confound strategy, and precision target. Receiver change without behavioral change supports only a local neural effect. Behavioral change without the receiver endpoint rejects the claimed sender-receiver path even if the stimulation has some other effect.

## 10. Frozen Replication Context

Discovery uses cue set A and its original left-right action mapping. Replication uses cue set B, a novel visual rendering of the same latent transition graph, with the left-right action mapping reversed. Replication data are collected only after the feature definitions, model families, preprocessing sensitivity set, SESOs, analysis code, and decision tables are frozen.

Model weights may be refit on replication training episodes, but the architecture families, hyperparameter budget, feature definitions, target, and evaluation code may not change. Stage A must clear the same primary comparison in context B. Stage B requires an independently localized sender-receiver pair using the same localization algorithm; the discovery phase angle and site are not transferred by hand. Context-specific estimates and intervals are reported even if a pooled hierarchical estimate is favorable. A pooled effect cannot conceal a sign reversal.

## 11. Confounds And Nulls

The design must address:

- common cue-locked input and evoked responses;
- movement and arousal;
- amplitude and aperiodic spectral changes;
- nonsinusoidal harmonics;
- reference and volume-conduction effects;
- trial order and slow drift;
- model-capacity and feature-count differences;
- phase-estimator and filter-edge artifacts; and
- perturbation sensation or nonspecific disruption.

Block-preserving label permutations, phase scrambling, long circular shifts, reference sensitivity, and the nonlinear raw-input comparator each destroy a different relation. They are reported as an ensemble, not as interchangeable evidence.

The nuisance plan is hierarchical. **Acquisition nuisances** include dropped channels, phase-lock failure, stimulation latency, reference drift, and artifact rejection. **Biological nuisances** include animal, session, site, arousal, movement, and baseline-excitability variation. **Task nuisances** include regime dwell, transition yield, learning carryover, cue imbalance, and order effects. **Model nuisances** include capacity, optimizer instability, preprocessing, feature count, calibration, and selection leakage. **Causal nuisances** include

dose mismatch, nonspecific stimulation, sensory effects, and failed target engagement. Each is assigned a measurement, exclusion rule, sensitivity range, or negative control before labels are opened.

## 12. Separate Decision Tables

### 12.1 Stage A observational decisions

| Result                                                                                                                          | Allowed conclusion                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Point estimate for $\Delta_A$ is at least $\delta_A$ , the 95 percent CI excludes zero, controls pass, and context B replicates | engineered phase clears the declared representation-utility threshold under these model classes; no information-creation or causal claim |
| 95 percent CI excludes zero, but the point estimate is below $\delta_A$                                                         | positive resolved gain below the declared practical threshold                                                                            |
| 90 percent CI lies entirely within $[-\delta_A, +\delta_A]$                                                                     | engineered phase is equivalent to no practically meaningful advantage under this lock                                                    |
| $M_{\text{raw}}$ or $M_{\text{dyn}}$ matches or wins, or gain fails controls or replication                                     | no demonstrated phase representation utility in this preparation                                                                         |
| Interval overlaps both zero and $\delta_A$                                                                                      | inconclusive; no positive or equivalence claim                                                                                           |

### 12.2 Stage B causal decisions

| Result                                                                                                                                                            | Allowed conclusion                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Point estimate for $\Delta_B$ is at least $\delta_B$ , the 95 percent CI excludes zero, matched-dose diagnostics pass, and the replication context agrees in sign | phase-specific causal residual supported for the declared receiver-transfer endpoint in this preparation    |
| 95 percent CI excludes zero, but the point estimate is below $\delta_B$                                                                                           | positive causal contrast below the declared practical threshold                                             |
| 90 percent CI lies entirely within $[-\delta_B, +\delta_B]$                                                                                                       | rejects a practically meaningful open-versus-closed receiver-transfer residual at this dose and preparation |
| Decoder gain without selective perturbation                                                                                                                       | observational representation utility only                                                                   |
| Perturbation contrast without decoder gain                                                                                                                        | causal receiver effect requiring a revised observational account; not retrospective validation of Stage A   |
| Interval overlaps both zero and $\delta_B$                                                                                                                        | inconclusive; absence of significance is not evidence of equivalence                                        |

### 12.3 No-result boundary

No row in either biological decision table has been reached. This preprint analyzes synthetic calibration data only; no biological data have been collected or analyzed, and no preregistration has been completed. A null Stage B result would not erase a Stage A predictive effect. A null result in either stage would not erase the dated SAN genealogy. Conversely, the genealogy cannot convert a null or inconclusive experiment into a positive result.

## 13. Relation To Micah’s SAN Genealogy

The implementation hypothesis is continuous with Micah’s documented SAN development, while the dates and source types remain explicit:

- an internally dated 2011 source and a publicly posted 2012 discussion develop predictive self / world modeling, recurrent feedback, and slower / faster oscillatory structure [12];

- the publicly posted 2017 Neural Lace Podcast develops spatial prediction, multisensory read / write, coincidence, and recurrent brainwave transfer [13];
- immutable 2022 Git history fixes the NAPOT, receiver-relative phase, distributed projection, and waveform / frequency formulations in public text [14]; and
- the SAN preprint and later papers turn those atoms into explicit comparator and perturbation proposals [15].

The source packet distinguishes four layers: the original recording and its native timestamp or immutable repository object; a clearer retranscription used as a semantic aid; older transcripts used only for discovery and version history; and later synthesis. Load-bearing quotations from audio require checking against the recording. No unclear older transcription is treated as quotation authority.

This chronology establishes the development of Blumberg’s proposal. It does not by itself establish biological truth, causal efficacy, or use by any named author, and it does not replace the experiment. Conversely, a null experiment would not erase the dated genealogy.

## 14. Conclusion

Prospective learning needs mechanisms that can be compared, not only metaphors about anticipating the future. SAN supplies one testable candidate: multiscale relative phase and receiver state may provide useful predictive bias for changing transitions and may provide a phase-specific control surface for rapid adaptation. This preprint turns that idea into three separable risks. First, the implemented synthetic pipeline distinguishes the four reference generators, but its weak-effect, high-noise, and intervention-artifact stress tests define substantial limits that must remain visible. Second, the representation claim must beat a nonlinear raw-waveform champion on held-out transitions and replication context. Third, the causal claim must survive a matched-dose open-versus-closed perturbation with a predeclared receiver endpoint and explicit identification assumptions. A behavioral claim additionally requires a specified path from receiver transfer to adaptation. If any comparison fails its SESOI, calibration, causal-assumption, or replication rule, the corresponding claim must be rejected or left inconclusive for the tested preparation. Passing the synthetic reference benchmark licenses evaluation of the pipeline; it does not license a claim that biological phase gating has been found.

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